

# Section 2.2: Solving Systems of Linear Equations by the Gauss-Jordan Method

Finite Mathematics MTH 132 Spring 2014

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# Solving Linear Systems by the Gauss-Jordan method

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$$\begin{array}{rcrcrcrcl} -2x & + & y & = & -8 \\ 2x & + & y & = & 4 \end{array}$$

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$$-2x + y = -8$$

$$2x + y = 4$$

$$\begin{array}{ccc} -2 & 1 & -8 \end{array}$$

$$\begin{array}{ccc} 2 & 1 & 4 \end{array}$$

step 1

Write the system in abbreviated form, called an *augmented matrix*.

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### step 2

Transform the matrix so that each column (except the last) has no more than one 0.

### step 3

Divide so the leading entry for each row is 1.



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### step 3

Divide so the leading entry for each row is 1.

### step 4

A row of 0's means no solution. Otherwise the last column gives the solution.

### Example 2.2.1.

Solve the system

$$\begin{array}{rclcl} -3x & + & y & = & 4 \\ 2x & - & 2y & = & -4 \end{array}$$

### Example 2.2.2.

Use the Gauss-Jordan method to solve the system.

$$x \quad \quad +z = 3$$

$$\quad y \quad -z = 1$$

$$x \quad +y \quad +z = 4$$

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$$x \quad \quad +z = 3$$

$$\quad y \quad -z = 1$$

$$x \quad +y \quad +z = 4$$

$$\left[ \begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & -1 & 1 \\ 1 & 1 & 1 & 4 \end{array} \right]$$

Problem 2.2.1. Use the Gauss-Jordan method to solve the system.

$$\begin{array}{rcrcrcrcl} x & + & y & = & 5 \\ 3x & + & 2y & = & 12 \end{array}$$

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Answer: (2,3)